

Paper 3.5

The Constant That Does Not Stay Constant*A Structural Audit of the Fine-Structure Measurement**Critique Companion to Paper 3 · Paper 3.5 of the Series*

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ABSTRACT

This is the critique companion to Paper 3, and it turns the inside-outside anatomy onto a single number the field treats as sacred: the fine-structure constant, measured at roughly one part in one hundred thirty-seven. The standard account holds this number to be an intrinsic, universal property of an isolated particle. This paper makes a narrower and harder claim than any in Paper 2.5, because it does not rest on the framework being right — it rests on the field's own published measurements. The number is not constant. It is measured at one value under gentle laboratory conditions and at a measurably different value when the apparatus drives higher energy through the same region. A quantity that changes when you change how hard the apparatus squeezes is not an intrinsic property of an isolated thing; it is a measure of how a medium responds to a load. This paper reframes the measurement accordingly: there is no empty space and no isolated particle — the apparatus forces geometry into geometry already present, the measured number tracks how much geometry is forced in, and its drift to a larger value at higher energy is the necessary consequence of adding more geometry to the equation, not the piercing of any unobservable screen. The same reframing dissolves the muon discrepancy — and exposes the manipulation behind it: the calculation was fitted to the electron, failed for the muon, and was rescued by a hand-fitted "hadronic" patch drawn from separate experiments and a non-derivable correction, even as the field's own two standard methods of computing it now disagree with each other. The paper then shows the same number shifting across three scales — cosmic (Webb's spatial dipole), material (the admitted shift in semiconductors), and atomic (the proton-radius discrepancy under a muon probe) — and exposes the double standard by which a physical medium is admitted in some cases and denied in others. It ties to the null-result cornerstone of Paper 2.5: the runs discarded as background are added geometry changing state as it seeks resolution, unrecognized only because the instruments are tuned for one expected signature. And it sets all of this against the documented unease of the field's own giants — Pauli, Feynman, Webb, and Barrow — who said on the record that this number is not understood and may not even be constant. The paper validates every recorded number and disputes only the inference drawn from them.

1. Purpose

Paper 2.5 audited a series of inferred particles and showed, case by case, that an entity was named where only a result was seen. This companion audits an inferred *property* — a constant said to belong to an isolated particle — and shows that the property is not constant at all, by the field's own measurements. The argument is deliberately built so that it does not depend on the framework being correct. It depends only on a fact the field publishes and a piece of reasoning anyone can follow: a number that shifts when the apparatus changes is not an invariant of the thing inside the apparatus.

The number is the fine-structure constant. The measurement is the anomalous magnetic moment of a trapped State 1 configuration — what standard physics calls the electron g-2 measurement in a Penning

trap. The claim under audit is that this measurement reads an intrinsic, universal constant inherent to an isolated particle in empty space.

2. The Apparatus Is Geometry Added to Geometry

The standard description of the measurement begins by calling the trap an isolated particle sitting alone in empty space. But there is no empty space to sit in. What the standard account treats as emptiness is, in this framework, geometry — the fabric itself, already present, already structured (Paper 2). The trap does not introduce a particle into a void; it introduces *more geometry* into geometry that was already there. A container of heavy, hard-bonded material is built, and a strong magnetic field is driven through the center of it, and in the framework a magnetic field is not an invisible fluid laid over nothing — it is a severe, directed addition of geometric structure, a compression of the fabric. The interior is not emptiness holding a particle; it is geometry, under heavy added load.

A State 1 configuration is then placed at the center of that load. Because a State 1 configuration is a localized looping structure *in* the fabric, it cannot be separated from the geometry it occupies, and now it sits in a region where the apparatus is continuously forcing in still more geometry. What happens next is the whole story of the measurement, and it is worth stating mechanically rather than as "orbit" and "wobble."

The added geometry from the apparatus is trying to resolve. That is what geometry does in this framework — it seeks resolution, back toward the photon baseline or into a settled state or a bond. But the geometry the apparatus forces in does not fit cleanly; much of it cannot resolve against the State 1 configuration as it stands. So it repels, and comes back, and repels again — each cycle a slightly different amount of unresolved geometry pressing in and being pushed off, each cycle a different value in the accounting. That repeated, never-quite-resolving exchange between the forced geometry and the configuration is exactly what the measurement records as the wobble. The wobble is not an isolated particle jittering in a void; it is the visible signature of added geometry failing to resolve, cycling in and out, computed slightly differently each pass.

And when, finally, the forced geometry does find the right reconciliation — when enough has been added in the right arrangement — the State 1 configuration changes state. That is not an anomaly or a destruction; it is the most ordinary thing in the framework. A configuration given enough geometry to resolve differently simply resolves differently. The measurement is not watching a free particle reveal an intrinsic property; it is watching added geometry seek resolution against a configuration under load, and eventually find it.

Figure 1. The Trap as a Vice

What is claimed: a free particle in empty space

What is actual: a configuration deformed under load

"isolated · undisturbed · intrinsic"

configuration

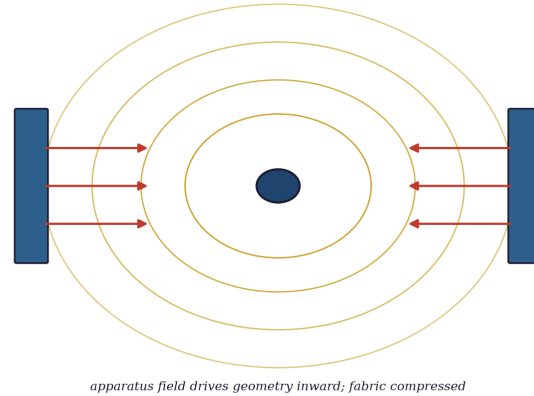


Figure 1. The trap as a vice: what is claimed to be a free particle in empty space is, in the framework, a configuration deformed under the apparatus's added geometric load.

3. The Circular Step

The inference proceeds in four moves, and the circularity is visible once they are laid out plainly. First, an intense artificial apparatus configuration aggressively deforms a State 1 configuration in the fabric. Second, the frequency of how that configuration resists the deformation is recorded. Third, that frequency is processed through a mathematical template that assumes from the start that the particle is isolated and the surrounding space is empty. Fourth, the number that comes out is declared a universal constant intrinsic to an isolated particle.

The conclusion is contained in the third step's assumption. The template presupposes the isolated particle in the empty void; the output is then read as proof of a property of that isolated particle. The apparatus that holds the vice shut never appears in the account. This is the same inference move audited throughout Paper 2.5 — a result produced by an apparatus, attributed to an entity the apparatus was never separated from — applied here to a property rather than a particle.

4. Why the Number Cannot Be a Constant: It Tracks the Geometry Put In

The decisive fact is not a framework claim. It is published. The value of the fine-structure number measured at gentle laboratory energies — the low-energy limit, near one part in one hundred thirty-seven — is not the value measured when the apparatus drives much higher energy through the region. At the energy scale of high-energy collider operation, the measured coupling is measurably larger, nearer one part in one hundred twenty-eight. The number the public is told is a fixed universal constant is, in the laboratory, a function of the energy the apparatus delivers — and the framework explains exactly why that must be so.

Here is the caveat that the standard account never states and that makes the whole drift inevitable: **the more energy the apparatus drives in, the more geometry is added to the equation.** Energy, in this framework, is not an abstract quantity turning a dial; it is added geometric structure forced into the region

(Section 2). A gentle measurement adds a little geometry; a high-energy measurement adds a great deal more. And the more geometry you force in, the more there is to disburse, the more that is seeking resolution at once, the more of it that cannot settle cleanly against the configuration under test. A larger amount of unresolved geometry pressing in and cycling produces a larger measured interaction strength. The number rises because there is simply *more geometry in the accounting*.

So of course the number changes with energy. It would be astonishing if it did not. The measurement is not reading an intrinsic property that happens to drift; it is reading how much geometry the apparatus put in, expressed as an interaction strength. At low energy, little added geometry, smaller number; at high energy, much added geometry, larger number. The value is dependent on the geometry forced into the test and is computed from it — which is the precise opposite of an invariant constant belonging to an isolated particle. The "constant" is a reading of the apparatus's own geometric load.

This is why the framework rejects the word "constant" for this number outright, and it does so without inventing anything. The standard account preserves the word by a different route: it holds that the particle is surrounded by an unobservable cloud that screens it at low energy and is pierced at high energy, so the higher number reveals a truer underlying charge. But there is no need for a cloud, a screen, or a bare charge. A number that rises as more geometry is added is just that — more geometry in the equation, more disbursement, a larger reading. One account invents an unobservable shield to protect a constant that refuses to stay constant; the framework reads the drift directly as the amount of geometry the apparatus introduced. The number is not a property of the particle. It is a measure of the geometry put in to make the measurement.

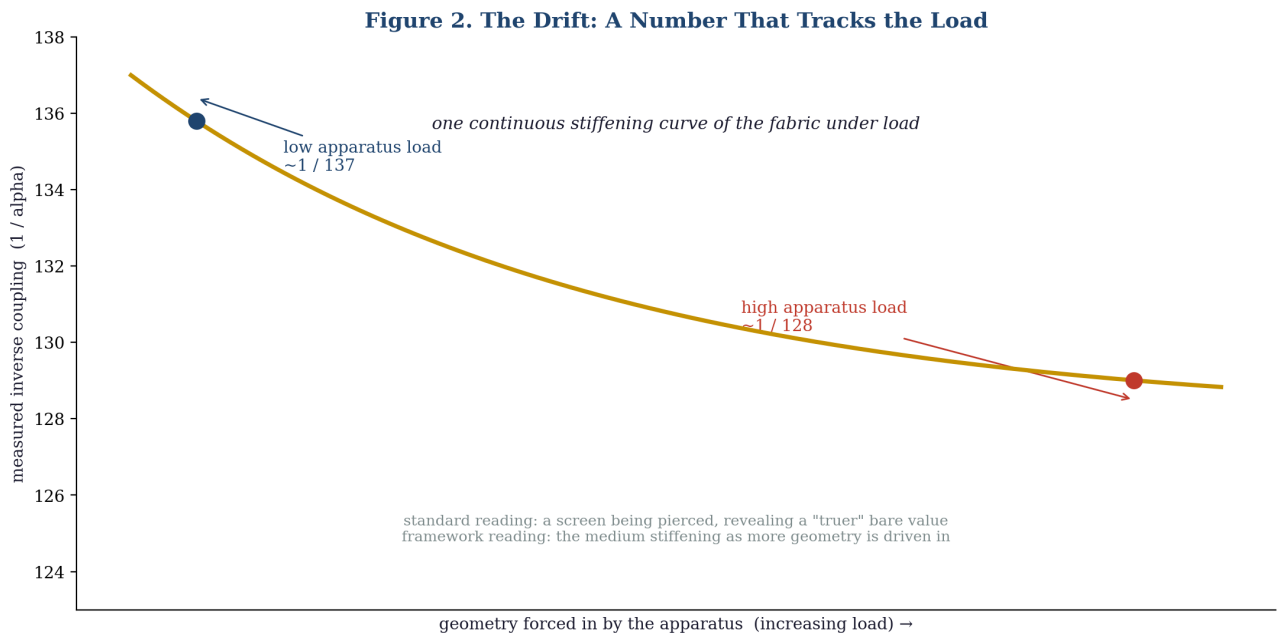


Figure 2. The drift: the measured coupling read at low apparatus load and at high apparatus load, as one continuous stiffening curve of the fabric rather than a screen being pierced.

5. The Muon, and the Shifting Ledger

5A. The Scale-Dependent Failure

The same template applied to a heavier configuration — the muon g-2 measurement — does not fit. The number that the calculation produces for the lighter configuration does not predict the heavier configuration's measured wobble. In the framework this is exactly what should happen: a different configuration places a different structural load on the fabric, and processes that fabric through a different scale of the outer temporal scan (Paper 3, Section 7B), so it meets the apparatus's forced geometry with a different response. There is no reason the response of one configuration should predict the response of another at a different scale. The failure is not a surprise to be patched; it is the expected behavior of a medium under a different load.

5B. The Arbitrary Ledger Split

What the standard account does next is the heart of this audit, and it must be stated precisely rather than waved at, because it exposes the manipulation directly.

The g-2 calculation is built as a power series in the fine-structure number. For the electron, the coefficients of that series were engineered over decades — computed from large families of Feynman diagrams and tuned until the series reproduced the baseline laboratory measurement to extraordinary precision. That is the first thing to see clearly: the equation was *fitted* to the electron. It was built to match the electron's measured wobble. Its agreement with the electron is therefore not an independent prediction confirmed by experiment — it is a template shaped, term by term, until it matched the one experiment it was made for.

Then the muon was placed in the same apparatus, and the fitted electron template failed. If the fine-structure number were a universal constant governing an isolated particle, the same series should have predicted the heavier configuration's wobble with no special pleading. It did not. And here is the move: rather than concluding that the template was only ever a fit to the electron, the field preserved it by changing the accounting method itself. They added a new class of terms to the muon's ledger that are not derived from the core theory at all — a "hadronic" contribution, justified by the claim that the heavier muon, unlike the light electron, is heavy enough to stir heavier unobservable structures out of otherwise empty space.

The decisive fact about this added term is that it cannot be computed from the theory it is rescuing. It is worth naming exactly what the patch is and where its numbers come from, because the specifics are more damning than any general charge. The added term is called the hadronic vacuum polarization contribution. It is obtained not from the core equations but by taking the measured cross-section of an entirely separate class of experiments — electron-positron annihilation into hadrons, supplemented by tau-lepton decay data — packaging that measured cross-section as the ratio quantity the field labels $R(s)$, and feeding it through a weighting integral into the muon's ledger as a "correction" sized exactly to close the gap. (The explicit form of that integral, and the full multi-tier split of the calculation, are laid out in Appendix A.) The structure of the move is therefore circular in the open: the template's failure is used to justify inventing a contribution the core theory cannot produce; the size of that contribution is then set by importing and curve-fitting data from different experiments until the prediction matches; and the forced match is reported as a triumph of precision. A clean, single template was used for the electron; a fractured, multi-tiered ledger with a hand-fitted patch imported from foreign experiments was required to force the muon's compliance. One particle, one set of rules; a heavier version of the same thing, a different set of rules bolted on. That is not a constant being confirmed across scales — it is an accounting method being rewritten whenever it fails.

5C. Their Own Two Methods Disagree

The clearest proof that this is bookkeeping rather than measurement is that the field's own standard calculations of the muon's wobble do not agree with each other. There are two roads to the same number. The data-driven (dispersive) method fits the hadronic patch from collision experiments, as Section 5B describes. The independent road is a supercomputer calculation from first principles — and the decisive detail is *how* that calculation has to be done. To compute the fabric's response without importing external collision data, the lattice method cuts spacetime into a discrete four-dimensional grid of points, a Euclidean lattice, and simulates the medium's response on that grid. In 2021 the Budapest-Marseille-Wuppertal (BMW) collaboration published exactly such a lattice result in *Nature*, and it pulled the prediction toward the measured value, undercutting the data-driven consensus. The two methods have remained in tension since; the field itself now refers to the standoff as "the new muon g-2 puzzle."

Two things follow, and both serve the framework. First, the deadlock itself is fatal to the picture of a clean universal constant: if the number were an invariant property of the particle, there would be one value, computed one way, agreeing with experiment — not two standard methods inside one theory disagreeing over the same quantity, so that the "constant" depends on which calculational filter the theorist lays over the data. Second, and more pointed: the only tool standard physics possesses that can approach the experimental value from pure first principles is a tool that explicitly treats space as a structured, discrete, responsive medium. The moment the calculation discretizes the so-called vacuum into a grid, the grid mechanics begin to account for the structural resistance the particle-in-a-void equations left out — which is why that road lands closer to reality. Their most precise first-principles instrument has to abandon the empty-void picture and model space as a medium to work at all. That is the framework's central claim, demonstrated by the standard model's own best calculation. The framework's reading needs none of the machinery: there is no universal number being approached by competing calculations, only the apparatus-dependent response of the fabric to the geometry forced into it, which naturally takes different values under different loads and at different scales. The standard model's internal disagreement is the framework's expected result, seen from inside the picture that cannot accommodate it. The detailed ledger of both methods is set out in Appendix A.

6. The Same Number Shifts Across Three Scales — and They Change the Rule Each Time

The drift from one part in one hundred thirty-seven to one part in one hundred twenty-eight inside a trap is not an isolated curiosity. The fine-structure number shifts at the cosmic scale, at the material scale, and at the atomic-probe scale — and at each scale the standard account adopts a different posture to protect the word "constant." Laid side by side, the three cases expose a single double standard: where the medium is admitted to be physical, the shift is accepted; where the medium is insisted to be empty, the shift is hidden behind unobservables. The framework reads all three the same way, because to the framework all three are the same thing — the fabric's response varying with its local load.

6A. The Cosmic Scale: Webb's Spatial Variation

Beginning around 1999, the astrophysicist John Webb and collaborators, including the cosmologist John Barrow, analyzed light from distant quasars passing through intervening gas clouds billions of light-years away, and reported that the fine-structure number appears to have taken a slightly different value in the distant past — and, in later work (2011), to vary with direction on the sky, tracing a preferred axis across the cosmos that has been described as a spatial dipole. The standard account treats the number as an

absolute scalar written identically into every point of empty space for all time, so a number with a geographic orientation is intolerable to it; the findings were accordingly attacked and sidelined, because the templates break if a "constant" can point in a direction.

The framework reads the dipole as exactly what it should be. If the number is not a property of an isolated particle but the elastic resistance of the continuous fabric to an axis tilt and an outer temporal scan, then large-scale structure in the fabric — regional differences in density and strain across cosmic distances — will make that resistance vary by region and direction. Webb's team did not measure mutating particles. They measured the large-scale directional variation of the fabric itself under cosmic loads. The dipole belongs to the geometry of space, not to the electron.

6B. The Material Scale: The Semiconductor Double Standard

On the laboratory floor the double standard is undeniable. In condensed-matter physics, when a State 1 configuration moves through a semiconductor or a quantum dot, physicists openly record that the effective value of the fine-structure number changes, and they state the reason without embarrassment: the surrounding atomic structure restricts the space and alters the local resistance. The medium is admitted to be physical, and the shift is accepted as the medium's doing.

But when the very same number shifts inside a trap — from one part in one hundred thirty-seven to one part in one hundred twenty-eight as the apparatus drives in more geometry — the same community refuses to say the obvious parallel thing: that the space in the trap is also a physical medium under strain. Instead it invents an unobservable cloud to keep the space "empty" and the particle "isolated." This is two contradictory rules for one phenomenon. When the medium is visibly made of atoms, a shifting number means the medium is acting on the configuration. When the medium is called empty space, the identical shifting number is forbidden from meaning the same thing, and a screen of unobservables is summoned to forbid it. The framework simply applies the honest rule — the one they already use for semiconductors — everywhere: the medium is physical, and the number shifts because the medium's load shifts.

6C. The Atomic-Probe Scale: The Proton Radius Puzzle

The third case is the one Paper 3 sets up directly through the scale-dependence of the temporal scan (Section 7B). In 2010, experimentalists measured the size of a proton using muonic hydrogen — replacing the usual light State 1 probe with a heavier configuration — and the proton came out measurably smaller than every electron-based measurement had said. The discrepancy threw the field into a decade-long crisis, addressed by adjusting error bars and re-examining constants rather than by questioning the assumption underneath.

That assumption is that the probe has no mechanical effect on the geometry it measures. The framework rejects it on principle. A heavier configuration places a different structural load on the fabric and processes it through a different scale of the outer temporal scan, so it deforms the local geometry differently than a lighter probe does. Measuring the same target with two probes of different load and getting two different sizes is not a paradox; it is the expected behavior of a responsive medium. The standard template failed because it assumed it was measuring a fixed sphere with a weightless ruler, when in fact each ruler presses on the fabric it measures. As the framework would put it: they tried to measure a trampoline with a bowling ball and a marble, and were astonished the shape came out different.

7. What the Discarded Data Actually Is: Geometry Changing State, Unwatched

This is the part of the test the standard account is blindest to, and it is the most beautiful part. When the apparatus forces geometry into the region (Section 2), that added geometry does what all geometry does: it tries to resolve — back toward the photon baseline, or into a settled State 1, or into a bond. In the course of trying, it passes through other states. A burst of added geometry can drive the configuration into State 3, into State 7, into whatever the local arrangement permits on the way to seeking resolution. These are real, ordinary state changes — exactly the kind of geometric transition Paper 3 describes.

But the instruments are not tuned to see them. The detectors are calibrated to trip on one specific signature — the resolved State 1 the experiment expects, or the specific fracture it is built to count. When the added geometry shifts into a State 3 or a State 7, or sheds a partial arrangement that matches no template, the apparatus does not recognize it as anything. It logs it as background, sub-threshold noise, or detector inefficiency, and discards it. The transition happened; the instrument simply was not listening on that channel.

That is the whole mechanism of the test laid bare. The apparatus forces geometry in; the geometry changes state trying to resolve; the instrument, tuned only for its expected signature, throws the rest away as noise. What is discarded is not nothing — it is the record of added geometry moving through states on its way to resolution, the very activity the experiment is unknowingly driving. A detector that recorded those discarded transitions would be watching the geometry resolve in real time. Instead it deletes them and keeps only the moments that match its template, then reports those matches as if they were the whole of what happened.

This is where the audit meets the cornerstone of Paper 2.5. The discarded runs are not absence; they are the most honest data in the experiment — the map of where the added geometry went when it was not doing what the instrument expected. A field that keeps only the matches and discards the transitions throws away the record of the geometry actually resolving, and then reads a "constant" off the narrow slice it kept. Publishing the discarded transitions alongside the kept matches would show the geometry changing state under load directly, with no unobservable shield required to hold a constant in place.

Figure 3. The Methodological Inversion



Figure 3. The methodological inversion: the standard chain discards null runs, imports unobservable corrections, and forces a constant; the framework keeps the null runs and reads a scale-dependent response.

8. The Number Was Never Settled — and the Field's Own Giants Said So

The claim that the fine-structure number is a settled, intrinsic property of an isolated particle is not even settled *within* the standard picture. Some of the most important physicists of the twentieth century stared at this number and said plainly that it was an unsolved mystery and a sign of an incomplete theory. This is public record, and it deserves to be named in full rather than gestured at, because it shows that the unease this paper formalizes has been sitting in the foundations of the field for nearly a century.

Wolfgang Pauli. Pauli — one of the founders of quantum mechanics, the author of the exclusion principle, a Nobel laureate — was, by the testimony of his colleagues, obsessed with the number 137 for much of his life. He regarded the fact that physics could measure the fine-structure constant in the laboratory and insert it into the equations, but could not derive *why* it takes the value it does, as evidence that the theoretical foundation was incomplete. He worked on the problem for years with the psychoanalyst Carl Jung, treating 137 as something close to a cosmic riddle. The detail that has passed into physics lore is the manner of his death: Pauli died in 1958 in a Red Cross hospital in Zürich, in room number 137 — the number he had chased his whole life. Whether one finds that eerie or merely a coincidence, the documented fact is that a founder of the theory spent a lifetime insisting this number was not understood.

Richard Feynman. Feynman — who built much of the very formalism, quantum electrodynamics, that is used to compute the electron's measured wobble, and won the Nobel Prize for it — was blunt about what the number meant. In his book *QED: The Strange Theory of Light and Matter* (1985) he wrote of the fine-structure constant that it has been a mystery since it was discovered, that all good theoretical physicists put this number up on their wall and worry about it, and that it is "one of the greatest damn mysteries of physics: a magic number that comes to us with no understanding by man." He described it as something "the hand of God" wrote, adding that "we don't know how He pushed His pencil." Feynman also called the renormalization procedure used to make these calculations work a "dippy process" and "hocus-pocus," and said outright that having to resort to it kept the theory from being proven self-consistent. The architect of the calculational machinery said, in print, that the number it produces is not understood and that the machinery itself was suspect.

John Webb and John Barrow. The unease is not only historical and not only theoretical. As set out in Section 6A, Webb, Barrow, and collaborators reported from 1999 onward that the fine-structure number appears to have differed in the distant past and to vary with direction across the sky. The finding remains contested — but the resistance it met is itself the tell: a number that varies with where and when it is measured cannot be a fixed, intrinsic property of an isolated particle, and the mainstream response was largely to dispute the data rather than follow its implication.

This paper does not enlist Pauli, Feynman, Webb, or Barrow as supporters of the three-axis framework. They worked within the standard picture and reached their own conclusions, and it would be dishonest to conscript them. The point in naming them — with dates, sources, and their own words — is narrower and devastating on its own terms: the field's own giants have said, on the record, that this number is not understood, that the machinery used to compute it is suspect, and that it may not even be constant. The framework's contribution is not to have *noticed* the number is strange. Pauli, Feynman, and Webb noticed. The framework's contribution is to say *what it is*: not a magic number written by the hand of God, but a reading of how much geometry the apparatus forces into the test — scale-dependent because the geometry is, and never constant because it was never a property of a particle to begin with.

9. What This Audit Validates and What It Disputes

The audit disputes none of the recorded numbers. The low-energy value was measured; the high-energy value was measured; the muon's response was measured; the discarded runs were recorded before they were discarded. Every measurement stands. What the audit disputes is the single inference laid over all of them: that a number which moves when the apparatus changes is an intrinsic, invariant property of an isolated particle in an empty void. The framework reads the same numbers as the response of a continuous fabric to a controlled, variable load — the baseline value, the drift with energy, and the elastic limit mapped by the quiet runs, all accounted for by one structural picture with no unobservable entities introduced.

The reflex this paper asks of the reader is the same two-part reflex Paper 2.5 installed, turned now onto properties as well as particles: when standard physics names a constant intrinsic to a thing, ask whether it stays constant when the apparatus changes — and ask what the discarded runs would have shown.

References

This paper is the critique companion to Paper 3 and part of the *Departure From the Photon Baseline* series. It cites the companion papers below by their published records, and the primary historical sources for the unease documented in Section 8.

1. Kadunic, D. (2026). *Departure From the Photon Baseline: A Universal Ruler for Mass, Gravity, and Time — The 27 Lagrangians of Gravity: A Survey of the Field's Standards of Evaluation* (Paper 1). Zenodo. <https://doi.org/10.5281/zenodo.20434286>
2. Kadunic, D. (2026). *The Three-Axis Framework and the Nine-State Classification of the 27 Geometric States* (Paper 2). Zenodo. <https://doi.org/10.5281/zenodo.20510019>
3. Kadunic, D. (2026). *They Assumed It; They Did Not See It: A Critique of Inferred Particles in Standard Physics* (Paper 2.5). Zenodo. <https://doi.org/10.5281/zenodo.20518514>
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6. Enz, C. P. (2002). *No Time to be Brief: A Scientific Biography of Wolfgang Pauli*. Oxford University Press. (Pauli's preoccupation with 137 and the circumstances of his death in room 137.)
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9. Aoyama, T., et al. (Muon g-2 Theory Initiative) (2020). "The anomalous magnetic moment of the muon in the Standard Model." *Physics Reports*, 887, 1–166. (The data-driven consensus prediction.)
10. Borsányi, S., et al. (BMW Collaboration) (2021). "Leading hadronic contribution to the muon magnetic moment from lattice QCD." *Nature*, 593, 51–55. (The first-principles lattice result that disagrees with the data-driven consensus.)
11. Pohl, R., et al. (2010). "The size of the proton." *Nature*, 466, 213–216. (The muonic-hydrogen measurement that opened the proton-radius puzzle.)

Paper 2 establishes the fabric, the photon baseline, the states, and the principle that what is called empty space is geometry; Paper 2.5 establishes the null-result cornerstone and the inference-audit method this

paper applies to a property rather than a particle; Paper 3 establishes the inside-outside anatomy and the state-change mechanics that Section 6 draws on directly.

A. A Structural Audit of the Anomalous Magnetic Moment Ledger

This appendix lays the standard g-2 calculation out as an accounting audit — its own equations, its own structure — and reads each tier through the three-axis framework. The purpose is not to dispute any arithmetic. Every number below is correctly computed within the standard picture. The purpose is to show what that picture has to do to keep a single constant standing, and to name where the ledger quietly imports its numbers from somewhere else.

A.1 The Fractionalization of the Baseline

Standard quantum electrodynamics calculates the anomalous magnetic moment of a configuration such as State 1 by treating the surrounding space as an empty void and adding a series of perturbative corrections, organized by the number of closed loops in their Feynman diagrams. Written as a power series in the fine-structure number, the moment takes the form

$$a(\text{QED}) = C_1 (\alpha / \pi) + C_2 (\alpha / \pi)^2 + C_3 (\alpha / \pi)^3 + C_4 (\alpha / \pi)^4 + C_5 (\alpha / \pi)^5 + \dots$$

where α is the fine-structure number (close to $1/137$) and C_1, C_2, C_3 , and so on are computed coefficients. To match the laboratory measurement, the standard calculation must be carried to fifth order in this series — the tenth order in the field's loop counting — which requires evaluating 12,672 distinct Feynman diagrams. That figure is not a framework claim; it is the documented count the field reports for the complete five-loop contribution.

The framework reads this infinite series as an accounting artifact forced by the wrong spatial template. Because the standard picture assumes an empty, non-responsive background, it has no way to represent a smooth continuous deformation of a medium directly; it must approximate that single physical curve as an unending sum of ever-smaller virtual events. In the framework the tiers are not populations of virtual particles blinking into existence. The baseline coefficient is the un-deformed geometric resistance of the three-axis fabric to the magnetic field the apparatus drives in; the higher-order coefficients are successive slices of the fabric's stiffening curve as the apparatus field compresses the local geometry harder. One continuous structural response, sliced into an infinite series because the template forbids naming the medium that is actually responding.

A.2 The Hadronic Vacuum Polarization: A Foreign Patch

For a heavier configuration — the muon — the power series above no longer matches the measurement, and the standard account closes the gap with a contribution it calls the hadronic vacuum polarization. The decisive structural fact is that this term cannot be computed from the core equations. It is built from imported data, through an integral of the form

$$a(\text{HVP}) = (1 / (3 \pi^2)) \left[\int_{(m_\pi)^2}^{\infty} K(s) / s R(s) ds \right]$$

where s is the center-of-mass energy squared, $K(s)$ is a smooth analytical weighting kernel, and $R(s)$ is an experimental cross-section ratio measured directly in electron-positron annihilation experiments — the rate at which those collisions produce hadrons — supplemented by tau-decay data. The structure is explicit in the equation itself: to compute a supposedly intrinsic property of one isolated configuration sitting in a

storage ring, the calculation copies a data ledger, $R(s)$, gathered from an entirely different class of high-energy collision experiments, and inserts it.

The framework reads the HVP integral as a direct, if unintended, confession of scale-dependent fabric displacement. A heavier configuration loads the local geometry more heavily, and as the apparatus drives in energy the added geometry shifts transiently through other states (a momentary State 3, a momentary State 7) on its way to seeking resolution, exactly as Paper 3 describes and as Section 7 of this paper develops. The $R(s)$ integral is a record of how the continuous medium redistributes geometry under a high, localized load. The standard account calls the same record "hadronic vacuum loops," but its own equation is integrating the elasticity curve of a physical medium.

A.3 The First-Principles Split: Dispersive versus Lattice

The present tension in the field follows directly from forcing a continuous medium into a flat-void template. There are two standard roads to the hadronic contribution. The dispersive (data-driven) method treats the vacuum as an empty canvas and manually injects the external $R(s)$ collision data to supply the missing response. The lattice method bypasses external collision data entirely and instead programs a supercomputer to compute the response by cutting spacetime into a discrete four-dimensional grid of points — a Euclidean lattice — and simulating the medium on it.

In 2021 the Budapest-Marseille-Wuppertal collaboration published a high-precision lattice calculation of the leading-order hadronic contribution in *Nature* that moved the theoretical prediction toward the Fermilab measurement and undercut the data-driven consensus, opening the standoff the field now calls the new muon $g-2$ puzzle. The structural verdict is the point that matters here, and it is not softened by which method ultimately prevails: the only tool standard physics has that can approach the measured value from pure first principles is a tool that explicitly treats space as a structured, discrete, responsive medium. The instant the calculation discretizes the vacuum into a grid, the grid mechanics begin supplying the structural resistance the particle-in-a-void equations omitted. The framework does not need to choose a winner in this dispute. It only observes that the standard model's most precise first-principles instrument has to model space as a medium to function — which is the framework's foundational claim, reached by the standard account's own hand.

A.4 On Renormalization

One feature of the machinery deserves to be named directly, because it is the step at which the standard calculation openly removes a quantity it cannot otherwise handle. In carrying the loop expansion out, the individual integrals do not yield finite numbers on their own; they diverge. The procedure that makes the calculation usable, renormalization, systematically absorbs those divergences into redefinitions of the theory's input quantities, so that the remaining finite part can be compared with experiment. The procedure is technically consistent and undeniably produces accurate predictions; that is not in dispute.

What is in dispute is the inference that a procedure built to subtract away infinities is evidence that the underlying picture — an isolated particle in an empty void — is the correct ontology. The framework reads the divergence as the symptom and the subtraction as the workaround: a continuous medium under load has a smooth, finite response, but a template that refuses to admit the medium and instead sums point-like virtual events in empty space will generate infinities at every order, which must then be removed by hand to recover the finite physical answer the medium would have given directly. The most economical statement of the framework's position is that the infinities are an artifact of the void template, and renormalization is the maintenance cost of keeping that template after it has stopped being physical. On

this point the framework does not stand alone, and Section 8 records that the architects of the formalism said as much in their own words: Feynman called the procedure a "dippy process" and "hocus-pocus" and stated that having to resort to it kept the theory from being shown self-consistent. The framework's only addition is to say what the medium is that would have made the subtraction unnecessary.

A.5 Consolidated Translation Ledger

The table maps the standard calculation's central terms to their framework readings.

Standard term	Standard definition	Framework reading
alpha (fine-structure number)	Intrinsic coupling constant of the electromagnetic interaction	Scale-dependent deformation ratio of the three-axis fabric under the apparatus's load (near 1/137 at baseline conditions)
QED loop expansion	Sum of virtual-particle loops screening a bare charge in empty space	Infinite-series approximation of one continuous geometric stiffening curve in a responsive medium
R(s) ratio integral (HVP)	Hand-fitted data integral tracking virtual hadron production in vacuum	Imported elasticity log: how the continuous fabric redistributes geometry under a high localized probe load
Renormalization	Systematic absorption of divergences into redefined inputs	Removal, by hand, of infinities that arise only because the void template refuses to admit the medium
Lattice discretization	Numerical scheme for computing QCD on a grid	The void template forced, in practice, to treat space as a structured responsive medium to reach the measured value

B. The Measurement Chain Behind the Trapped Value

This appendix examines how the baseline value is obtained in a trap, in order to locate precisely where the assumption of a fixed, intrinsic constant enters. The aim is to separate what is measured from what is assumed.

B.1 The Derived-Mass Assumption

The mass of the trapped configuration is not weighed. It is inferred from the cyclotron resonance frequency f_c at which the configuration circulates in a magnetic field B , through the standard relation $f_c = qB / (2 \pi m)$, solved for m . This is a derivative determination: the apparatus measures a frequency response to an applied field and converts it to a mass under the explicit assumption that the mass is a fixed constant. Every variation that appears in the measurement is then attributed to the field, never to the quantity assumed constant. If the configuration's resistance to inversion is in fact a variable state of the surrounding fabric rather than a fixed scalar, this single assumption routes that variability into the wrong column of the ledger from the outset.

B.2 The Trap-Potential Dependence

To hold the configuration in place, the apparatus applies a trapping potential, and that potential places the surrounding region of fabric under a standing strain. The observed resonance is therefore a function of two things, not one: the configuration's own geometry and the load the trap imposes on the local medium. In the framework's reading, changing the trap potential changes the strain on the fabric and shifts the observed resonance. The standard account holds the medium to be empty and so has nowhere to assign that contribution; it reads the entire resonance as a property of the configuration alone. This is the

mechanism behind the drift the body of the paper describes, from roughly one part in one hundred thirty-seven toward one part in one hundred twenty-eight as the apparatus drives in more geometry: the number tracks the load, because the load is part of what is being measured.

B.3 A Closed Calibration Loop

The final structural feature is that the measured value is also used to calibrate the apparatus that measures it. When the equipment is tuned, the expected value of the number is among the standards the tuning is referenced against. A system in which the output is used to set the input cannot, by construction, return a result that contradicts the expected output: deviations are read as the apparatus being out of calibration and are tuned away. This does not require any bad faith on the part of the experimenter; it is a structural property of a self-referential measurement chain. The consequence is simply that the stability of the number across such measurements is partly enforced by the calibration procedure rather than discovered independently of it, and a measurement chain calibrated against a value it then reports cannot serve as evidence that the value is an invariant of nature rather than a fixed point of the apparatus.

C. Why the Electron and Muon Measurements Are Not Commensurate

The claim that one fine-structure number governs both the electron and the muon rests on comparing two measurements made under geometrically incompatible conditions, then mathematically transforming one to match the other.

The electron value is obtained in a static, low-energy trap: a configuration held near equilibrium, its resonance read at stable frequency. The muon value comes from a high-energy storage ring: a configuration carried at relativistic speed under extreme angular load, its properties read from a decaying population in flight. These are not the same measurement performed on two masses. They are two different geometric environments — one near rest, one under severe kinematic strain — and the load each places on the surrounding fabric is correspondingly different.

To bring the two into a single ledger, the standard account applies relativistic transformations that map the high-energy environment onto the low-energy baseline, correcting for the clock difference between a stationary trap and a circulating ring. Within the standard picture this is routine and correct. But it is a normalization, not an independent agreement: the muon environment does not return the electron's number on its own; it is transformed until it does, and the residual mismatch is then assigned to additional contributions, including the imported hadronic patch of Appendix A. In the framework's reading the residual mismatch is not an anomaly to be absorbed but the expected signature of two different loads on a responsive medium. A heavier configuration processed through a different scale of the outer temporal scan (Paper 3, Section 7B) deforms the fabric differently and meets the apparatus's forced geometry differently; there is no reason its response, once honestly compared, should reduce to the lighter configuration's. The framework treats the muon as a distinct geometric configuration of the same medium rather than a heavy copy of the electron, and on that reading the "anomaly" is not a gap to be patched but a measurement of the medium responding to a different load.

This appendix follows, step by step, what the standard account does when a measurement does not match the prediction. The point is not to allege bad faith. It is to lay the documented sequence side by side so the reader can see the pattern for themselves: at each stage where the data resisted the constant, a new term was added rather than the assumption questioned. The sequence is a matter of published record; the

reader is left to judge what it amounts to.

D.1 The Pattern, Stated Plainly

When the fitted electron template failed to predict the muon's measured wobble (Section 5), the standard account had two roads open to it. One was to ask whether the number was a universal constant at all. The other was to keep the constant and add a new contribution sized to absorb the difference. The field took the second road, and it is worth being precise that this was a choice, not a necessity: the discrepancy could equally have been read as evidence that the number tracks the configuration and its load, which is the framework's reading. Instead, a "hadronic vacuum polarization" term was introduced (Appendix A.2), justified by the proposal that the heavier muon stirs heavier unobservable structures out of the surrounding space — structures that the lighter electron does not. The term cannot be derived from the core theory; its magnitude is set by importing data from separate collision experiments and fitting it until the prediction matches.

The structural observation is simple and does not require imputing motive: a contribution whose existence is proposed because it is needed, and whose size is then fixed by tuning it to the gap it was proposed to fill, has not been independently confirmed. It has been fitted. When such a term is later reported as a successful prediction of the theory, the success is partly built in. This is not a claim that anyone falsified anything; it is a claim that a fitted parameter cannot also serve as independent evidence for the framework that needed it.

D.2 When the Patch Itself Splits in Two

The clearest sign that something other than measurement is happening appears when the patch is computed two different ways and the two ways disagree. As Section 5C and Appendix A.3 set out, the hadronic contribution can be obtained either by importing collision data (the dispersive method) or by simulating the medium on a discretized grid (the lattice method, e.g. the BMW 2021 calculation). These two roads to the same quantity have produced different values, and the field is openly divided over which to trust. A genuine intrinsic constant does not come in two values depending on which calculational route the theorist selects. A quantity that does is being constructed, not read off nature — and the framework names the missing variable that both methods are circling: the response of the medium to the load, which neither void-based bookkeeping captures cleanly and which the grid method partially recovers precisely because it abandons the void.

D.3 The Standing Option of a New Term

The deeper structural feature is that the standard template can, in principle, absorb any future discrepancy the same way: by proposing a further contribution and fitting its size to close the new gap. A framework with an open-ended supply of fittable correction terms is difficult to falsify, because there is always another term available when the data resists. This is not an accusation; it is a statement about the logical structure of the method. The framework offered in Paper 3 takes the opposite stance deliberately: it commits in advance to a bonding-compatibility matrix that forbids specific outcomes (Paper 3, Section 8), so that a single stable composite violating the matrix would break it. A theory that can be broken by one measurement and a theory that can absorb any measurement by adding a term are doing different things, whatever the precision each reports.

The strongest version of this paper's argument does not require the framework to be correct. It requires only that the same phenomenon be explained the same way wherever it appears. This appendix lays the field's own treatments of the fine-structure number side by side and asks one question of each: is the

surrounding medium admitted to be physical, or insisted to be empty? The answer changes from case to case — and it changes in the direction that protects the constant each time.

E.1 The Same Shift, Read Two Ways

In condensed-matter physics, when a State 1 configuration moves through a semiconductor or a quantum dot, the effective value of the fine-structure number changes, and the field states the reason without controversy: the surrounding atomic structure alters the local response (Section 6B). The medium is visibly made of atoms, so a shifting number is read, correctly, as the medium acting on the configuration.

Inside a trap, the very same number shifts as the apparatus drives in more energy — from roughly one part in one hundred thirty-seven toward one part in one hundred twenty-eight. Here the surrounding region is called empty space, and the identical shift is not allowed to mean the medium is acting on the configuration. Instead an unobservable screening cloud is introduced to keep the space empty and the particle isolated. Two physically identical observations — a number that shifts with its surroundings — receive two incompatible explanations, and the deciding factor is whether the medium is something the eye already accepts as physical. The framework's correction is the most conservative move available: apply the rule the field already uses for semiconductors everywhere, and the unobservable cloud is no longer needed.

E.2 The Same Probe Problem, Renamed a Puzzle

The proton-radius case makes the inconsistency sharper still (Section 6C). When the proton was measured with a heavier probe (muonic hydrogen) it came out smaller than every lighter-probe measurement had reported, and the field named the contradiction "the proton-radius puzzle" and spent a decade adjusting error bars around it. The assumption never examined was that the probe has no mechanical effect on what it measures. A measurement that gives two sizes for one object under two different probe loads is, in any other setting, a sign that the probe is part of the system. The framework reads it that way directly: a heavier configuration loads the fabric differently and reads a different size, with no puzzle remaining.

E.3 The Pattern Across Scales

Set the three scales side by side and the consistency test fails in one direction every time:

Scale	Observation	When medium is admitted physical	When medium is called empty
Material (semiconductor)	number shifts with surroundings	shift accepted as the medium's doing	—
Laboratory (trap, rising energy)	number shifts with apparatus load	—	shift attributed to a screening cloud being pierced
Cosmic (quasar lines)	number varies with direction/epoch	—	finding disputed; a directional constant called intolerable
Atomic (muonic vs electronic probe)	object measures two sizes	—	named a "puzzle," assumption left intact

The framework's claim is the entry that fills every empty cell with the same answer: the medium is physical in all four rows, the number tracks its local load in all four, and no unobservable entity is required in any of them. The standard account is not wrong to admit the medium for semiconductors. It is only inconsistent in refusing to admit it everywhere the same signature appears. That inconsistency — not any imputed motive — is the finding of this appendix. A constant that must be protected by a different rule at each scale was

not a constant to begin with.